

Fitting strategy of resonance curves from microwave resonators with non-idealities.

Nicola Pompeo*, Kostiantyn Torokhtii*, Fabio Leccese[†], Andrea Scorza[‡], Salvatore Sciuto[‡], Enrico Silva*[§]

*Dipartimento di Ingegneria, Università Roma Tre, Via Vito Volterra 62, 00146 Roma, Italy.

[†]Dipartimento di Scienze, Università Roma Tre, Via della Vasca Navale 84, 00146 Roma, Italy.

[‡]Dipartimento di Ingegneria, Università Roma Tre, Via della Vasca Navale 79, 00146 Roma, Italy.

[§]Email: enrico.silva@uniroma3.it

Abstract—Microwave resonators are of widespread use in technological applications as well as in many research fields. In all cases, as microwave filters or as material characterisation devices or as sensors, their use requires the accurate measurement of their quality factor Q and resonance frequency ν_0 , which are the main parameters shaping their response. The most accurate measurements are obtained through a full determination of all the complex scattering coefficients S_{ij} of a two-port coupled resonator, through a Vector Network Analyzer, and subsequent fitting in the complex plane. These approaches allow also to compensate for resonator non-idealities which alter the resonator response and thus impact on Q and ν_0 accuracies. On the other hand, cost effective and fast measurement setups rely on power sensors which allow scalar measurements only. In these cases, the resonator ideal response takes the shape of a lorentzian curve. It is therefore of interest to optimise the fitting strategy of scalar measurements in order to obtain the better fit goodness and robustness against perturbations. In this work we study the most common sources of perturbation and their models, in order to assess which fitting strategy yields the best results when the perturbation cause is not known a-priori. We perform both numerical simulations and tests on actual experimental data. We find that the fit model describing a cross-coupling between two resonator ports is the most robust approach.

I. INTRODUCTION

Electromagnetic resonators are ubiquitous devices used both in technological applications [1]–[3] and in many research fields [4]–[6]. Their working principle requires that an oscillating electromagnetic field, confined by conducting walls and/or dielectric boundaries and having a specific spatial configuration (mode), achieves a balance between stored electric and magnetic energies for a specific resonant frequency ν_0 . At resonance, the ratio between the stored electromagnetic energy and the dissipated energy per oscillation cycle defines the resonator quality factor Q . When inserted within a microwave line through one or two coupling ports (for reflection or transmission operation), the resonator presents a bell-shaped response centered around its resonant frequency and with a bandwidth inversely proportional to Q . Thanks to this property, various types of microwave filters can be made [7]. As a tool for material characterization, a resonator properly loaded with conducting, dielectric or magnetic samples, allows to determine the complex resistivity, permittivity or magnetic permeability of the materials, respectively [1], through the measurement of its parameters Q and ν_0 . In these situations, typical required accuracies are, respectively, 10 for $Q \sim 10^4$

and ~ 1 –10 ppm for ν_0 . This principle is also largely exploited in sensors for quantities like humidity, temperature or strain [8] which use the permittivity dependence of a sensing dielectric material on the quantity to be measured. In general, the resonator quality factor Q and resonance frequency ν_0 are obtained through a fit of the frequency (ν) dependent resonator scattering coefficients $S_{ij}(\nu)$, measured around the resonant frequency. The measurements of the complex S_{11} , S_{22} and S_{21} , which require a Vector Network Analyzer (VNA), and their combined fitting, yield the lowest measurement uncertainty [9]. The latter is true also considering commonly encountered non-idealities in the resonator response which occur in real setups. In this framework, the search for accurate and optimized fit algorithms is a long standing [10], [11] and still very active [12]–[14] subject. On the other hand, cost-effective measurement apparatuses rely on power sensors and diode-based microwave sources to measure the scalar power output of the resonator, and hence the squared modulus of a scattering coefficient $|S_{ij}|^2$. In these cases, usually the transmission scattering $|S_{21}|^2$ only is considered. In this way, it is possible to neglect the contribution of the microwave line to the resonator response, apart from a residual scale factor which affects the resonance peak value S_M only, generally of no interest.

In the following we focus on the latter scenario, concerning cost-effective scalar measurements of $|S_{21}|^2$, with the aim to determine the fitting method most effective and robust against the most common non-idealities exhibited by real resonators.

The manuscript is organized as follows. In Section II, the problem will be formalized, describing the most common non-idealities that will be addressed, and the fitting models described. In Section III, numerical simulation results will be presented in order to assess the capabilities of the proposed fits, while in Section IV two applications to case studies, given by real measurements performed with 8 GHz and 11 GHz resonators will be discussed. Finally, conclusions will be drawn.

II. THE PROBLEM

The response of a transmission-operated resonator is, in ideal conditions, the following [12], [15]:

$$S_{21}(\nu) = S_{res}(\nu; \nu_0, Q_L, S_M) = \frac{S_M}{1 + 2iQ_L \frac{\nu - \nu_0}{\nu_0}} \quad (1)$$

where $S_M = S_{21}(\nu_0)$ is the $|S_{21}(\nu)|$ peak (real) value, $Q_L = Q/(1 + \beta_1 + \beta_2)$ is the loaded quality factor of the resonator with β_1 and β_2 the coupling coefficients of the two resonator ports. For the common undercoupled setup, $\beta_1 \sim \beta_2 \ll 1$ so that $Q_L \simeq Q$, which will be used in the following. The square modulus $|S_{21}(\nu)|^2$ of the ideal response function yields a lorentzian curve:

$$|S_{21}(\nu)|^2 = \frac{S_M^2}{1 + 4Q^2 \left(\frac{\nu - \nu_0}{\nu_0} \right)^2} \quad (2)$$

whose fit requires the three *real* parameters S_M , Q and ν_0 . When non-idealities come into play, the model of the resonator response has to be changed accordingly. Apart from the omnipresent signal noise, which determines an additive random uncertainty on the S_{21} frequency points, the most common and hence here considered perturbations to the ideal response are a cross coupling between the resonator ports and a spurious mode close to the main resonator mode. Both of them qualify as systematic sources of uncertainty. A cross coupling between the two ports of the resonator can be described through an additive complex contribution S_c to $S_{21}(\nu)$. Given the narrow frequency range of measurement around ν_0 , the frequency dependence of S_c can be neglected with respect to $S_{res}(\nu)$ so that S_c is a complex constant [9], [12]. Hence the corresponding fitting model becomes:

$$S_{21}(\nu) = \frac{S_M}{1 + 2iQ \frac{\nu - \nu_0}{\nu_0}} + S_c \quad (3a)$$

with

$$S_c = S_{cR} + iS_{cX} \quad (3b)$$

so that $|S_{21}|^2$ requires five real fitting parameters.

The other common perturbation of the ideal resonance curve is given by a spurious mode, centered at ν_{0s} , which determines an additional resonant contribution $S_{res,spurious}(\nu) = S_{res}(\nu; \nu_{0s}, Q_{Ls}, S_{Ms})$ in $S_{21}(\nu)$:

$$S_{21}(\nu) = \frac{S_M}{1 + 2iQ \frac{\nu - \nu_0}{\nu_0}} + S_{res,spurious}(\nu) \quad (4a)$$

with

$$S_{res,spurious}(\nu) = \frac{S_{Ms}}{1 + 2iQ_{Ls} \frac{\nu - \nu_{0s}}{\nu_{0s}}} \quad (4b)$$

Within this model, the fit requires six real fitting parameters. Actually, since the peak (i.e., the frequency ν_{0s}) of the spurious mode is usually outside of the measured frequency range, only a “tail” of this second curve contributes. Hence $S_{res,spurious}$ can be approximated to a frequency dependent term $S_{tail}(\nu)$ defined through two parameters only. The overall fitting model is:

$$S_{21}(\nu) = \frac{S_M}{1 + 2iQ_L \frac{\nu - \nu_0}{\nu_0}} + S_{tail}(\nu) \quad (5a)$$

with

$$S_{tail}(\nu) = -i \frac{S_{Ms}}{2Q_{Ls}} \frac{1}{\frac{\nu}{\nu_{0s}} - 1} \quad (5b)$$

which depends on five real parameters only (the three main parameters, plus the ratio Q_{Ls}/S_{Ms} and ν_{0s}).

In Fig. 1(b), black line and symbols, the lorentzian curve for an ideal resonance, computed from Eq. 2, is reported together with the skewed curves as obtained in the cases of cross (Eq. 3) or spurious mode (Eq. 4) perturbations. In Fig. 1(a), an experimentally measured resonance is shown: it can be seen both the presence of noise and the skewing of the resonance curve.

In the following, the aim is to determine which is the best fitting strategy (i.e. use of the ideal, cross or spurious-mode models) against a non-ideal resonance affected by either a cross coupling or spurious mode perturbation, without knowing the actual nature of the perturbation.

III. NUMERICAL SIMULATIONS

In this Section, we discuss the efficacy and robustness of the various fit models, presented previously, against different test scenarios. The process consists of two steps. First, we numerically compute a resonance curve which incorporates either a cross perturbation (Eq. 3) or a spurious mode perturbation (Eq. 4). In order to keep contact with real measurements, we add a gaussian white noise (SNR $\sim$ 35 dB) to the computed curves. Then we fit the numerically obtained curve with the various models: the “cross” model of Eq. 3, the “full” (Eq. 4) and “tail” (Eq. 5) models for the spurious mode. The main mode parameters are taken among the commonly encountered values in microwave experiments: an example of real world resonance curve is provided in Fig. 1(a). The resonance frequency and maximum value are kept fixed at $\nu_0 = 10$ GHz and $S_M = 2.5 \cdot 10^{-2}$ ($|S_M|^2 \sim -30$ dB), respectively. The Q factor is varied around 10^4 , the typical value of medium- Q resonators, in the 2000–20000 range in order to detect possible dependencies of the results from the resonance curve width (the higher the Q , the narrower the resonance). The frequency point number is $\sim 10^5$ to ensure the smoothness of the simulations. This allows to avoid to dynamically adjust the fitting frequency range, as often done with the acquisition frequency range during measurements, since it can result in an additional “jitter noise” to the resulting fitting parameters.

The parameter space for the cross and spurious mode perturbations is vast, even by requiring the maximum value of the perturbing contribution to be at least one order of magnitude lower than S_M . Hence, in the following we report and discuss selected typical cases. First, in order to test the behaviour of the various models against a balanced scenario, we select two perturbations (cross coupling and spurious mode) which produce similar skewing of the resonance curve, despite their difference in the analytical expressions. This choice is depicted in Fig. 1(b) (numerical parameters are reported in the caption). In this balanced situation, as it can be expected, both the cross fit (Eq. 3) and spurious fit models (Eq.s 4 and 5) produce almost identical results (not shown here) when applied either to a cross or to a spurious mode perturbation. It is therefore checked that, despite the presence of the noise, (i) each model

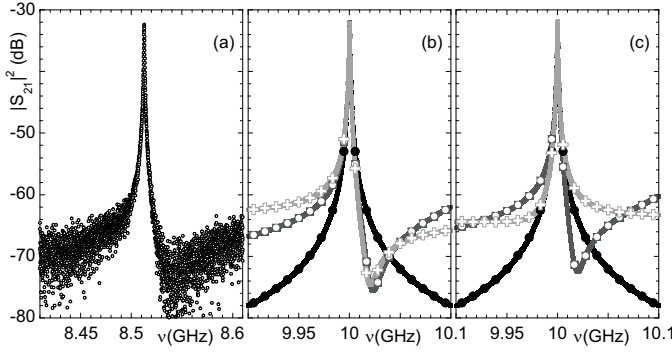


Fig. 1. Panel (a): Experimentally measured skewed resonance. Panels (b) and (c): numerically computed skewed resonance curves (without added noise): cross-coupling perturbation (crosses), spurious mode perturbation (squared open circles), and corresponding ideal resonance reported for reference (full dots). Main mode parameters are $Q = 10^4$, $\nu_0 = 10$ GHz and $S_M = 2.5 \cdot 10^{-2}$ (see text). In panel (b), the perturbation parameters are chosen to produce similar skewed curve, despite the different analytical forms: $S_c = (-2 + i6) \cdot 10^{-4}$ and $\nu_{0s} = 10.2$ GHz, $Q_{Ls} = 100$ and $S_{Ms} = 2 \cdot 10^{-3}$. In panel (c), the perturbation parameters are chosen to produce quite different skewed curve: $S_c = (6 - i2) \cdot 10^{-4}$ and $\nu_{0s} = 10.2$ GHz, $Q_{Ls} = 80$ and $S_{Ms} = 2 \cdot 10^{-3}$.

is actually capable of fitting the correspondingly computed curve and (ii) the two models are at least equivalent when the skewed resonance does not allow to recognise the nature of the perturbation. We now consider a different test scenario, depicted in Fig. 1(c). The perturbation parameters (whose values are reported in the figure caption) have been altered with respect to the previous sets so that, leaving unchanged their absolute weights $|S_c|$ and S_{Ms} , they produce significantly different skewings of the resonance curve.

To discuss the results of the fitting, we introduce the normalized deviation of a quantity m with respect to the corresponding true nominal value m_t , as $(m - m_t)/m_t$. The normalized deviations of Q , ν_0 and S_M are reported as a function of the nominal Q factor in the panels (a)-(c) of Figs. 2 and 3, for the cross perturbation and spurious mode respectively. In panels (d) the fit residuals, computed as customarily in the least squares method:

$$residual = \sum_{i=1}^n \frac{(|S_{21,data,i}|^2 - |S_{21,fit,i}|^2)^2}{n} \quad (6)$$

are reported. Here, $S_{21,data}$ and $S_{21,fit}$ represent the data values to be fitted and the computed fit values, respectively, n is the number of frequency points and $\sum_i$ is the sum extended over all the frequency points.

We now discuss the obtained results. In Fig. 2 it can be seen that when the perturbation is due to a cross coupling, it is better fitted, as expected, by the corresponding fit model expression (Eq. 3). Indeed, the latter yields visibly better results in terms of the resonance frequency, whereas the other fits (ideal and spurious mode) are affected by errors up to 3 ppm for the resonance frequency. The above error of the fit is reflected also by the fit residuals, which consistently indicate a better overall fit obtained through the cross model. Moreover, the errors of the fits obtained by the other models are more relevant for low

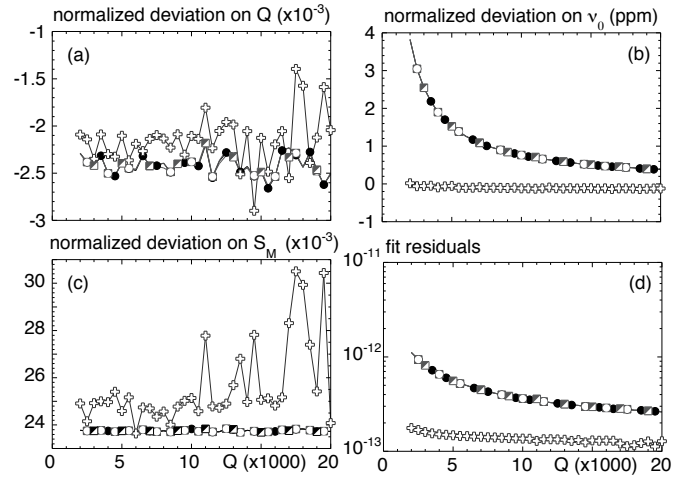


Fig. 2. Fitting results of a cross coupling perturbed resonance. Panels (a)-(c): normalized deviation of the fitting parameters from the corresponding nominal ones, as a function of the nominal Q factor. Panel (d): fit residuals. Symbols are as in Fig. 1; in addition, half-filled squares used for the spurious mode “tail” fit model. A subset of the data points is shown to avoid crowding.

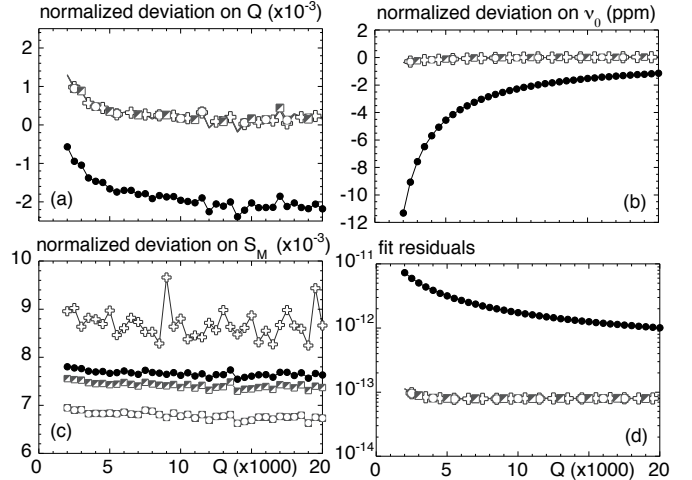


Fig. 3. Fitting results of a spurious mode perturbed resonance. Panels (a)-(c): normalized deviation of the fitting parameters from the corresponding nominal ones, as a function of the nominal Q factor. Panel (d): fit residuals. Symbols as in Fig. 2. A subset of the data points is shown to avoid crowding.

nominal Q values, as those which can be obtained in resonator loaded with very dissipative materials and/or with geometries different from cylindrical [16], [17]. In terms of Q and S_M , on the other hand, there are no significant differences.

Taking now into consideration a perturbation due to a spurious mode, Figure 3 shows that, differently from the previous case, the cross coupling and the spurious mode models yield similar results, while the standard three parameter model is definitely inaccurate in terms of both Q and ν_0 . These observations are consistent, again, to the fit residual values. The cross-coupling fit yields results slightly worse only for S_M which, as already stated, is usually of no interest.

Finally, considering the results of all the four scenarios here presented (Figs. 2 and 3), it can be noted that the five

parameters, spurious mode, “tail” expression of Eq. 5 gives the same results as the six-parameters, complete expression of Eq. 4.¹ Thus, as a secondary result of this work, the effectiveness of the simpler approximated expression here proposed is validated.

Summarizing, the cross model qualifies itself as a robust fit model, unaffected from the actual unknown origin of the resonance curve perturbation, and thus capable of yielding accurate estimates of the desired resonator parameters Q and ν_0 . In order to further validate the present results, in the following Section we take into consideration real, experimental resonance data on which we evaluate the behaviour of the various fit models.

IV. APPLICATION TO EXPERIMENTALLY MEASURED DATA

In this Section we take into consideration experimental resonance data taken with different microwave dielectric resonators [6], [16], [17] in various setups, selected in order to cover a wide range of Q factor values and, possibly, different perturbations. In the first setup, a cylindrical dielectric resonator [6], operated in transmission, is used. One of the bases has been partially covered by a flat sample of Nb, which superconducts below the critical temperature $T_c = 7.4$ K [18], [19]. To test the proposed models, the choice of a superconducting sample allows to experimentally explore a wide dynamics in Q values, in the medium-high range.

The resonator is excited in the TE_{011} mode with $\nu_0 \sim 8.2$ GHz. The microwave currents induced in the sample volume cause additional dissipation and reactive energy storage as a function of the material complex resistivity, thus contributing to the resonant mode Q and ν_0 [6]. The resonance curves $|S_{21}(\nu)|^2$ around the resonance frequency are acquired at different temperatures T varied around the superconductor critical temperature T_c , to keep track of the sample superconducting transition (inset of Fig. 4(a)). Qualitatively, by going below T_c a reduction of the power dissipation on the sample is expected, which corresponds to an increase of the resonator Q . Similarly, the change in the London penetration depth [20] of the superconductor below T_c determines a change of the sample reactance and ultimately of ν_0 .

The fit parameters Q and ν_0 obtained through standard (Eq. 1), cross (Eq. 3), spurious full (Eq. 4) and spurious tail models (Eq. 5) are reported as a function of T in Fig. 4, panels (a) and (b). The corresponding fit residuals, computed according to Eq. 6, with $|S_{21,data}|^2$ given by the measured data values, are reported in panel (c) of the same Figure. It can be seen that the fitted Q values are essentially the same, independently from the fit model used.

On the other hand, the resonant frequency ν_0 values show some discrepancies. In particular, the cross fit values are different from the values yielded by the other fit models. This is particularly evident above T_c (i.e., at low Q values), where the relative difference between the cross fit and the other fit

¹This fact is consistent with a preliminary observation of significant cross-correlations between parameters in the full, six-parameters spurious mode fit.

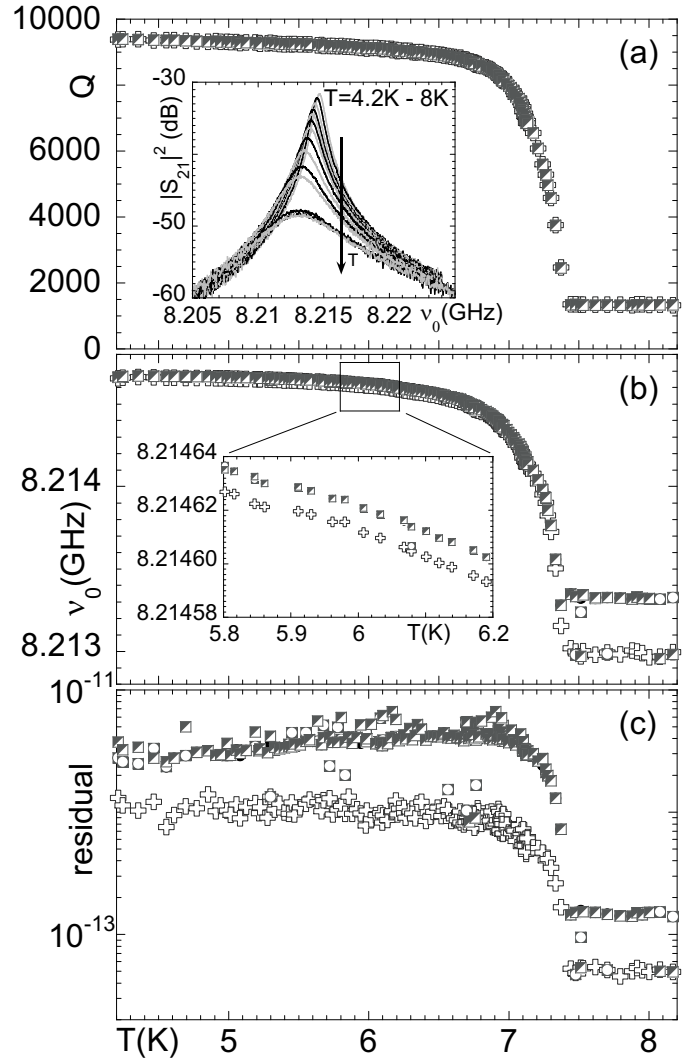


Fig. 4. Fits through various models of the experimental resonance curves taken by varying the temperature T . Panels (a)-(c): Q fit values, ν_0 fit values and (c) fit residuals, according to Eq. 6. Inset of panel (a): selected resonant curves, acquired at varying T , as measured through a dielectric resonator incorporating a superconducting Nb sample. An abrupt variation in the resonance curves is present at ~ 7.4 K. Fit symbols are as in Fig. 2. Inset of panel (b): zoom of panel (b) for a specific T range, highlighting a discrepancy of ~ 1 ppm between cross fit model and the others models.

values is as high as ~ 40 ppm. This difference goes down to ~ 1 ppm at 6 K, temperature at which the resonator exhibits higher Q values.

This is consistent with the results of the numerical simulation study presented in the previous Section, according to which the larger discrepancies between the different fit models (i) concerned the resonant frequency and (ii) appeared at low Q . Considering panel (c) of Fig. 4, it can be seen that the cross fit residuals are consistently lower than the residuals of the other fit models, at all T .

To explore the low- Q behaviour we used different setups, namely rectangular prism dielectric resonators (RDR) made of high [16] (resonator RDR1) and moderately high [17] permittivity (resonator RDR2). These devices are designed to

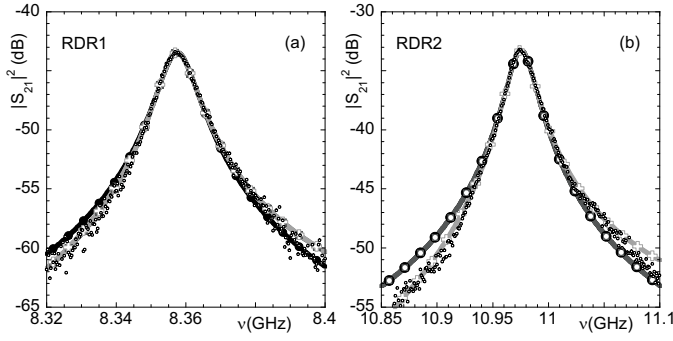


Fig. 5. Fits through various models of the experimental resonance curves measured through two distinct rectangular dielectric resonators RDR1 (a) and RDR2 (b) (see text). Experimental data points are reported as small open symbols; fit symbols are as in Fig. 2. The measurements shown are taken along the direction of the samples maximum electrical conductivity.

be able to detect and measure the anisotropy in the resistivity of flat conductive samples, so that their overall sensitivity (i.e. their Q values) has been correspondingly subjected to a trade-off. Hence, Q values between 400 and 800 are shown. Both resonators are operated in transmission and excited in the TE_{011} mode, with resonant frequencies ~ 8.31 GHz and ~ 10.9 GHz, respectively. The flat conductive sample, made of anisotropic pyrolytic graphite, is measured at room temperature along the directions of maximum conductivity. The experimental resonance curves for both RDRs are reported in Fig. 5, panels (a) and (b), along with the fits performed through the standard, cross and spurious-tail models (the spurious-full model fit is not shown, since it is essentially identical to the spurious-tail one). In Table I, the fit parameters Q and ν_0 , and the fit residuals, are reported.

It can be seen that, in this low- Q scenario, the cross model yield a slight (see the fit residuals) but appreciable improvement in the fit quality, visible in the reproduced asymmetry of the experimental resonance curve with respect to the standard ideal model. In overall, in the entire Q range experimentally tested, the cross-coupling model yields an appreciable improvement with respect to the standard three-parameter fit. With respect to the alternative models, mainly the spurious ones, it is either on-par or yields better results. Hence, consistently with the considerations of the previous Section, the values obtained through the cross fit can be considered the most trustworthy and accurate also in the analysis of experimental data.

V. CONCLUSIONS

We have studied different fit models in order to identify the best model to extract a resonator quality factor Q and resonance frequency ν_0 from its resonance curve, measured from the squared modulus of its transmission coefficient, as typically done in cost-effective, scalar measurement setups. We have shown that when the curve is skewed by superimposed perturbations, the fitting with an ideal lorentzian curve with the addition of a cross-coupling perturbation term is the best-behaving approach.

TABLE I
FIT PARAMETERS AND RESIDUALS.

Measurement	Fit model	Q	ν_0 (GHz)	fit residuals
RDR1	standard	785	8.357490	$9.7795 \cdot 10^{-14}$
	cross	787	8.357278	$6.8889 \cdot 10^{-14}$
	spurious-tail	787	8.357276	$6.8886 \cdot 10^{-14}$
RDR2	standard	433	10.975385	$1.6570 \cdot 10^{-11}$
	cross	436	10.974639	$5.5733 \cdot 10^{-12}$
	spurious-tail	433	10.975385	$1.6570 \cdot 10^{-11}$

In particular, this five-parameter fitting model ensures the best results when compared with the traditionally used simpler approaches, the three parameter ideal lorentzian fit or even the three-point 3 dB rough-and-dirty Q estimation. On the other hand, when compared with other, alternative five-parameter fit models, it shows up as the most robust even if the perturbation arises from other sources. This result has been obtained through the analysis of extensive numerical simulations and further validated analysing real world experimental data. In particular, it has been shown that the fitted resonance frequency is the quantity most affected by inaccurate fit models, with possible deviations up to ~ 100 ppm, to be compared to required accuracies, in typical sensor and material characterization setups, ~ 1 – 10 ppm. Due to the vastness of the parameter space that can be addressed through the numerical approach, a limited number of selected scenarios has been considered in the here presented simulations. The extension of the results to very low Q (below 10^2) can be of interest to passive temperature sensors [21] and low-frequency small volume antennas [22]. We defer to further studies the exploration of a larger part of the parameter space, including the effect of different noise levels on the fits accuracy and the effects of two or more spurious modes. As far as the experimental validation is concerned, further refinements can be envisaged with the evaluation of the statistical errors of the fitting parameters.

REFERENCES

- [1] L. F. Chen, C. K. Ong, C. P. Neo, V. V. Varadan and Vijay K. Varadan, "Microwave Electronics: Measurement and Materials Characterization", ed. Wiley, 2004.
- [2] E. Fratticcioli, M. Dionigi, and R. Sorrentino, "A Simple and Low-Cost Measurement System for the Complex Permittivity Characterization of Materials", *IEEE Trans. Instrum. Meas.*, vol. 53, pp. 1071–1077, Aug. 2004.
- [3] B. Kapilevich and B. Litvak, "Microwave measurements of dielectric properties using a gap-coupled multi-mode coaxial resonator", *Measurement*, vol. 48, pp. 21–28, Feb. 2014.
- [4] J. Sheen, "Study of microwave dielectric properties measurements by various resonance techniques", *Measurement*, vol. 37, pp. 123–130, Mar. 2005.
- [5] J. Krupka, "Contactless methods of conductivity and sheet resistance measurement for semiconductors, conductors and superconductors", *Meas. Sci. Technol.*, vol. 24, Art. no. 062001, Apr. 2013.
- [6] N. Pompeo, K. Torokhtii, E. Silva, "Dielectric resonators for the measurements of the surface impedance of superconducting films", *Meas. Sci. Rev.*, vol. 14, pp. 164–170, Jun. 2014.
- [7] G. L. Matthaei, L. Young, and E. M. T. Jones, "Microwave filters, impedance-matching networks, and coupling structures", Dedham, MA: Artech House Books, 1980.

- [8] T. Aftab, A. Yousaf, J. Hoppe, S. Stoecklin, T. Ostertag, and L. Reindl, "A parallel plate dielectric resonator as a wireless passive strain sensor" *Sensors Appl. Symp. (SAS), IEEE*, pp. 1–6, 2015.
- [9] K. Leong and J. Mazierska, "Precise measurements of the Q factor of dielectric resonators in the transmission mode-accounting for noise, crosstalk, delay of uncalibrated lines, coupling loss, and coupling reactance," *IEEE Trans. Microw. Theory Tech.*, vol. 50, pp. 2115–2127, Nov. 2002.
- [10] D. Kajfez, "Numerical Data Processing of Reflection Coefficient Circles," *IEEE Trans. Microw. Theory Tech.*, vol. MTT-18, pp. 96–100, Feb. 1970.
- [11] D. Kajfez, "Linear Fractional Curve Fitting for Measurement of High Q Factors," *IEEE Trans. Microw. Theory Tech.*, vol. 42, pp. 1149–1153, Jul. 1994.
- [12] P. J. Petersan and S. M. Anlage, "Measurement of resonant frequency and quality factor of microwave resonators: Comparison of methods," *J. Appl. Phys.*, vol. 84, pp. 3392–3402, Sep. 1998.
- [13] C. Deng, M. Otto, and A. Lupascu, "An analysis method for transmission measurements of superconducting resonators with applications to quantum-regime dielectric-loss measurements," *J. Appl. Phys.*, vol. 114, Aug. 2013, Art no. 054504.
- [14] S. Probst, F. B. Song, P. A. Bushev, A. V. Ustinov, and M. Weides, "Efficient and robust analysis of complex scattering data under noise in microwave resonators," *Rev. Sci. Instrum.*, vol. 86, Feb. 2015, Art no. 024706.
- [15] D. Kajfez, "Dielectric resonators", second edition. SciTech Publishing, 1998.
- [16] K. Torokhtii, N. Pompeo and E. Silva, "A rectangular dielectric resonator for measurements of the anisotropic microwave properties in planar conductors", *Meas. Sci. Technol.*, vol. 25, 2014, Art no. 025601.
- [17] N. Pompeo, K. Torokhtii and E. Silva, "Design and test of a microwave resonator for the measurement of resistivity anisotropy," *Measurement*, vol. 98, pp. 414–420, Feb. 2017.
- [18] E. Silva, N. Pompeo, S. Sarti, "Wideband microwave measurements in Nb/Pd₈₄Ni₁₆/Nb structures and comparison with thin Nb films", *Supercond. Sci. Technol.*, vol. 24, Jan. 2011, Art no. 024018.
- [19] N. Pompeo, K. Torokhtii, C. Cirillo, A. V. Samokhvalov, E. A. Ilyina, C. Attanasio, A. I. Buzdin and E. Silva, "Thermodynamic nature of the $0 - \pi$ quantum transition in superconductor/ferromagnet/superconductor trilayers", *Phys. Rev. B*, vol. 90, Aug. 2014, Art no. 064510.
- [20] C. P. Poole Jr., H. A. Farach, R. J. Creswick and R. Prozorov, "Superconductivity", Third Edition, Elsevier, 2014.
- [21] H. Cheng, X. Ren, S. Ebadi, Y. Chen, L. An and X. Gong, "Wireless Passive Temperature Sensors Using Integrated Cylindrical Resonator/Antenna for Harsh-Environment Applications", *IEEE Sensors Journal*, vol. 15, no. 3, pp. 1453–1462, Mar. 2015.
- [22] Oleksiy S. Kim, "Low-Q Electrically Small Spherical Magnetic Dipole Antennas", *IEEE Trans. Antennas Propag.*, vol. 58, n. 7, pp. 2210–2216, Jul. 2010.